

RESEARCH ARTICLE | AUGUST 07 2025

Flow through packed beds: Experimental and numerical validation of the Quinn fluid flow model

M. Nikitin ; H. M. Quinn ; D. Pashchenko  



Physics of Fluids 37, 084113 (2025)

<https://doi.org/10.1063/5.0283733>



Articles You May Be Interested In

A new semi-empirical model for pressure drop prediction in a packed bed of smooth and rough particles: Derivation and verification

Physics of Fluids (January 2026)

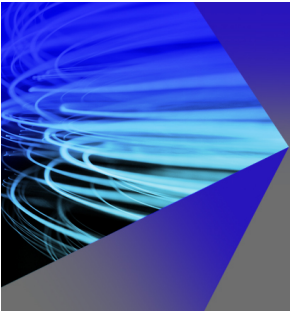
Flow through randomly packed beds of rough particles: Model prediction and experimental validation

Physics of Fluids (October 2025)

Exhaust mixing noise computed from time-accurate simulation of compressible free jet turbulence

J. Acoust. Soc. Am. (September 1998)

17 April 2026 07:46:40



AIP Advances

Why Publish With Us?



21DAYS
average time
to 1st decision



OVER 4 MILLION
views in the last year



INCLUSIVE
scope

[Learn More](#)



Flow through packed beds: Experimental and numerical validation of the Quinn fluid flow model

Cite as: Phys. Fluids **37**, 084113 (2025); doi: [10.1063/5.0283733](https://doi.org/10.1063/5.0283733)

Submitted: 3 June 2025 · Accepted: 5 July 2025 ·

Published Online: 7 August 2025



View Online



Export Citation



CrossMark

M. Nikitin,¹ H. M. Quinn,² and D. Pashchenko^{1,3,a)}

AFFILIATIONS

¹Department of Mechanical Engineering, Guangdong Technion—Israel Institute of Technology, 241 Daxue Road, Shantou, Guangdong 515063, China

²Department of Research and Development, The Wrangler Group LLC, Brighton, Massachusetts 02135, USA

³Faculty of Mechanical Engineering, Technion—Israel Institute of Technology, Technion, Haifa 3200003, Israel

^{a)}Author to whom correspondence should be addressed: dmitry.pashchenko@gtit.edu.cn

ABSTRACT

This study presents the experimental and numerical validation of the Quinn fluid flow model (QFFM), a novel theoretical framework for predicting pressure drop in packed beds and closed conduits. The QFFM, derived from first principles, establishes a universal linear relationship between the normalized dimensionless pressure gradient (P_Q) and fluid current (C_Q), expressed as $P_Q = k_1 + k_2 C_Q$. The model bridges the gap between empirical correlations and particle-resolved simulations, offering an approach for both empty and particle-packed systems. Experimental validation was conducted using a recirculating flow loop with precise pressure and temperature measurements, while numerical simulations employed high-fidelity computational fluid dynamics model. Numerical tests were conducted for a wide range of operational and design parameters of the packed beds for two types of fluid: water and air. Results demonstrate excellent agreement between QFFM predictions and experimental data across laminar, transient, and turbulent flow regimes, with discrepancies below 3.7%. The QFFM outperforms traditional models like the Ergun equation by inherently accounting for tortuosity and microscale flow phenomena. This work highlights the model's potential for optimizing industrial packed-bed systems, providing a useful tool for engineers and researchers.

© 2025 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>). <https://doi.org/10.1063/5.0283733>

I. INTRODUCTION

Packed beds are widely utilized in various engineering applications, including chemical reactors, filtration systems, and heat exchangers, due to their high surface area-to-volume ratio and efficient mass and heat transfer characteristics.^{1,2} The ability to accurately predict fluid flow behavior and pressure drop through these systems is important for optimizing design and operational efficiency.³ However, the complex, tortuous pathways formed by randomly packed particles introduce significant challenges in modeling pore-scale hydrodynamics, making precise predictions difficult.⁴

A key challenge in analyzing flow through packed beds lies in accurately capturing the pressure drop across the bed, which is influenced by factors such as particle size and its shape, porosity, fluid velocity, and particle roughness.^{5,6} Several equations and models have been developed to predict pressure drop in packed beds, each with varying levels of complexity and applicability: Ergun equation,⁷ Blake-Kozeny equation (laminar flow regime),⁸ Burke-Plummer equation (turbulent flow regime),⁹ Forchheimer equation, Macdonald modified

Ergun equation.¹⁰ Wu and Hibiki provided an overview of various correlations for pressure drop prediction in a packed.¹¹ The variety of equations is explained by the fact that the pressure drop depends on many parameters.

Traditional models, such as the Ergun equation, provide a semi-empirical framework but often fail to account for microscale flow phenomena, including local velocity gradients and inertial effects at high Reynolds numbers. In addition, the Ergun equation has low accuracy in the prediction of pressure drop for low D/d ratio (where D and d is tube and particle diameter, respectively).^{12,13} These limitations can lead to discrepancies between predicted and experimental results, particularly in systems with nonspherical particles or polydisperse packings.

Computational fluid dynamics (CFD) modeling is widely used to predict flow characteristics in fixed beds.^{14–16} However, simulating a fixed bed numerically involves significant computational costs. This is because even test problems may contain hundreds of particles in the packed bed, resulting in computational meshes with tens of millions of elements. Despite advances in computational fluid dynamics (CFD)

and experimental techniques, existing models still exhibit gaps in their ability to fully describe the intricate flow dynamics within packed beds. For instance, the Ergun equation assumes a uniform flow distribution and does not explicitly consider variations in pore geometry or particle orientation. Such simplifications can result in significant errors when applied to real-world systems where heterogeneity plays a dominant role.

There are many papers for the prediction of pressure drop in the packed beds. The main approach for those papers is to derive a semi-empirical equation based on the Ergun equation or other fundamental models. Macdonald *et al.*¹⁰ published a paper evaluating two friction factor-Reynolds number correlations—the nondimensional Forchheimer equation¹⁷ and a modified Ergun equation⁷—using extensive experimental data. They concluded that the Forchheimer equation's physical basis is robust, while the modified Ergun equation, though less rigorous, predicts results for unconsolidated media within $\pm 50\%$ accuracy. The study emphasizes the importance of correcting for nonspherical particle shapes and using proper averaging methods for particle size. Empirically, they found that replacing the Ergun equation's porosity term (ε^3) with $\varepsilon^{3.6}$ significantly improves fit accuracy across a wide porosity range (0.36–0.92). The authors affirm that the Forchheimer–Ahmed model's two-term (viscous + inertial) approach remains valid, though simplified channel-based models suffice for unconsolidated media.

Koekemoer and Luckas extended the Ergun equation to industrial-scale packed beds of large, polydisperse coal, char, and ash particles, addressing limitations of the original equation derived from small, monodisperse spheres.¹⁸ The authors demonstrate that material type (coal/char/ash) and particle size distribution (PSD) significantly influence bed porosity and sphericity, with ash beds exhibiting the highest porosity due to particle roughness and shape. Crucially, PSD width impacts porosity more than particle diameter, as finer particles fill gaps between larger ones, reducing interstitial space. Through regression analysis, new Ergun constants were derived for each material (e.g., 77.4 and 2.8 for coal), outperforming the classical values (150 and 1.75) in predicting pressure drops for composite beds resembling industrial gasifiers. The findings underscore the need for material-specific correlations in industrial reactor design.

Buchwald *et al.* reported that the Ergun equation can accurately predict the pressure drop, adding adjustable constants.¹⁹ The study challenges the common perception of the Ergun equation as a purely empirical model with adjustable constants (k_1 , k_2), arguing instead that its deviations for rough, irregular particles stem from inadequate measurement of specific surface area. Through analysis of 3D (three dimensional) particle morphology (via x-ray microscopy), the authors demonstrate that dynamic image analysis—a 2D (two dimensional) method—fails to capture both macroscale shape and microscale roughness, leading to inaccurate surface area estimates. The paper asserts that the Ergun equation's foundational physics remains valid, with discrepancies arising primarily for coarse, rough particles (e.g., clinker), where resolution limits ($\sim 10\ \mu\text{m}$) and unaccounted fractal surface effects skew results. Notably, it highlights tortuosity as an overlooked parameter embedded in k_1 and k_2 , suggesting future work to quantify its influence. The findings reject *ad hoc* adjustments to Ergun's constants, advocating instead for improved surface characterization across scales. This work reframes the debate by emphasizing rigorous parameter determination over empirical fitting.

Although many researchers have proposed semi-empirical models that describe pressure losses in packed beds accurately enough, Quinn proposed a different approach and derived the Quinn fluid flow model (QFFM).²⁰ In his study, Quinn introduced a fundamental principle derived from first principles, which establishes a linear relationship between the normalized dimensionless pressure gradient (P_Q) and fluid current (C_Q) for fluid flow in closed conduits, expressed as $P_Q = k_1 + k_2 C_Q$. The law applies universally to both empty conduits and those packed with solid particles, with tortuosity effects inherently incorporated in its normalization framework. Quinn proposed that under near-static flow conditions, the relationship reduces to a constant k_1 , while k_2 determines the slope under increasing fluid acceleration. Quinn's work provides a unified theoretical framework that bridges empty and packed-bed flow dynamics, offering significant potential for fluid dynamics modeling and industrial applications.

In this study, the authors present experimental and numerical validation of the QFFM across a broad range of operating and design parameters. The primary objective is to provide a derivation of the model and to demonstrate that the QFFM—by bridging the gap between empirical correlations and particle-resolved simulations—serves as a practical tool for engineers and researchers working with packed bed systems.

II. QUINN FLUID FLOW MODEL

The full derivation of the Quinn fluid flow model is provided in Quinn's previous paper.²⁰ QFFM is based on the following principles:

- (1) The QFFM is focused on pressure-driven fluid flow in closed conduits, which may or may not contain solid particles.
- (2) The model accommodates, seamlessly, both varieties of conduits, meaning it is not relevant whether the conduit contains solid obstacles or is devoid of same.
- (3) All conduits in the context of the QFFM are considered "packed conduits" which means that the conduits contain a discrete number of particles, n_p , which may or may not contain a solid skeleton. A conduit such as a capillary, for instance, which contains no solid obstacles, is considered a packed conduit containing particles of free space.

Based on these principles, the QFFM definition of a packed bed is based upon the basic concept of porosity, as that term applies to both the particles which make up the packed bed, i.e., particle porosity ε_p , and the packed bed itself, i.e., packed bed porosity ε . Additionally, the particles themselves are also defined in terms of their average spherical particle diameter equivalent d_p . However, since the operation of any conduit under study must conform to the laws of continuity, i.e., the conservation laws, particle diameter d_p , and conduit porosity ε , are not independent variables. In other words, the value of the one defines the value of the other. Accordingly, a packed bed in the context of the QFFM is defined in terms of three particle elements, n_p , d_p , and ε_p , and two packed bed variables, n_{pq} , the volume of the conduit expressed in terms of number of particles of diameter d_p and the porosity of the packed conduit ε . Thus, four variables, n_p , n_{pq} , ε_p , and ε , are dimensionless quantities which constitute the concept of term definition reciprocity mentioned above and are both necessary and sufficient to define the milieu in a packed bed appropriate for a discussion of the hydrodynamics of the fluid flow regime.

For purposes of examining the relationship in a packed bed between the driving force ΔP and the resulting fluid volumetric flow rate, q , the QFFM quantifies a packed bed in terms of its specific anatomy. This is accomplished by defining the packed conduit porosity ε as a mathematical function of n_p and ε_p , including several subsets of ε , namely:

- conduit particle volume fraction

$$(1 - \varepsilon_0) = n_p/n_{pq},$$

- conduit particle skeletal volume fraction

$$\varepsilon_{sk} = (1 - \varepsilon_p)n_p/n_{pq},$$

- conduit external volume fraction

$$\varepsilon_0 = (1 - n_p/n_{pq}),$$

- conduit internal volume fraction

$$\varepsilon_i = \varepsilon_p(n_p/n_{pq}),$$

- conduit total volume fraction

$$\varepsilon_t = 1 - (1 - \varepsilon_p)n_p/n_{pq}.$$

Accordingly, having now established a mathematical framework to accommodate a discussion of fluid flow within the packed bed, the QFFM introduces the concept of the hypothetical Q channel (HQC) in terms of its characteristic dimension, i.e., its diameter $d_c = d_p/\text{abs}(1 - \varepsilon_0)$, which is the ratio of the spherical particle diameter equivalent and the magnitude of the particle volume fraction.

The QFFM is based upon a concept of fluid flow that is more than 100 years old.²¹ The underlying theory is systematically derived from that concept, consisting of a universal relationship between fluid flow rate, q , and differential pressure, ΔP , using 10 sequential development steps. These steps are as follows:

- (1) Pressure drop in a general form:

$$\frac{\Delta P}{L} = aq + bq^2, \quad (1)$$

where a and b are empirical coefficients which balance the equation.

- (2) Forchheimer equation:

$$\frac{\Delta P}{\rho_f g L} = a_{\text{For}} \mu_s + b_{\text{For}} \mu_s^2, \quad (2)$$

where $\frac{\Delta P}{\rho_f g L}$ represents hydraulic gradient; $a_{\text{For}}, b_{\text{For}}$ represent Forchheimer coefficients; ρ_f represents fluid density; g represents acceleration due to gravity; and μ_s represents fluid superficial velocity.

- (3) Substitute for a_{For} and b_{For} :

$$\frac{\Delta P}{\rho_f g L} = \frac{4r_h^3 \pi \delta \eta \mu_s}{3 \rho_f g d_c^2} + \frac{\delta^2 \lambda \mu_s^2}{2 \pi g d_c}, \quad (3)$$

where $r_h = 4$, $\delta = 1/\varepsilon_0^3$, η is fluid absolute viscosity, and λ is wall normalization coefficient.

- (4) Rearrange by multiplying across by $\rho_f g$:

$$\frac{\Delta P}{L} = \frac{268 \delta \eta \mu_s}{d_c^2} + \frac{\delta^2 \lambda \rho_f \mu_s^2}{2 \pi d_c}. \quad (4)$$

- (5) Judicious grouping of terms:

$$\frac{\Delta P}{L} = 268 n_v + \frac{\delta \lambda n_k}{2 \pi}, \quad (5)$$

where n_v, n_k are viscous hydraulic force and kinetic hydraulic force exerted per unit element of fluid control volume, respectively; $n_v = \frac{\delta \eta \mu_s}{d_c^2}$; $n_k = \frac{\delta \rho_f \mu_s^2}{d_c}$.

- (6) Normalize for viscous terms by dividing across by n_v :

$$\frac{\Delta P}{n_v L} = 268 + \frac{\delta \lambda n_k}{2 \pi n_v}. \quad (6)$$

- (7) Identify the Q-modified Ergun equation:

$$f_v = A + B \text{Re}_m, \quad (7)$$

where $f_v = \frac{\Delta P}{n_v L}$, which represents viscous friction factor; $A = 268$, which represents viscous constant; $B = \frac{\delta \lambda}{2 \pi}$, which represents kinetic constant; and $\text{Re}_m = \frac{n_k}{n_v}$, which represents modified Reynolds number. It should be noted that the fluid motion term is the product of two entities: n_v , the viscous hydraulic force exerted per unit element of fluid control volume and $(4 \pi r_h^3/3 = 268)$ also known as the “viscous constant,” where r_h is the ratio of surface area to cross-sectional area of a spherical particle, i.e., $r_h = 4$. Incidentally, we point out that the value of 270 for the viscous constant in the Kozeny/Blake equation was identified as early as 1965 by J.C. Giddings.²²

- (8) Normalize for r_h (hydrodynamic drag) by dividing across by $r_h = 4$,

$$\frac{f_v}{4} = 67 + \frac{\delta \lambda \text{Re}_m}{8 \pi}. \quad (8)$$

- (9) Identify the universal flow equation (Quinn’s law):

$$P_Q = k_1 + k_2 C_Q, \quad (9)$$

where $P_Q = \frac{f_v}{4}$, which represents fluid-drag normalized dimensionless pressure gradient; $k_1 = \frac{64 \pi}{3} \approx 67$, which represents universal viscous constant; $k_2 = \frac{1}{8 \pi} \approx 0.04$, which represents universal kinetic constant; and $C_Q = \delta \lambda \text{Re}_m$.

- (10) Rearrange terms to identify the impact of wall normalization coefficient (λ) via the dimensionless permeability (Θ) of the HQC:

$$\Theta = \frac{Q_N}{P_Q} = \frac{1}{\frac{k_1}{Q_N} + \frac{\lambda}{2 \pi r_h}}, \quad (10)$$

where $Q_N = \delta \text{Re}_m$, which instantaneous fluid current.

Based on the mathematical transformations described in detail in the paper,²⁰ we obtain an expression for determining the pressure drop in the packed bed

$$\frac{\Delta P}{L} = \frac{A(1 - \varepsilon_0)^2 \mu_s \eta}{\varepsilon_0^3 d_p^2} + \frac{B(1 - \varepsilon_0) \rho_f \mu_s^2}{\varepsilon_0^3 d_p}. \quad (11)$$

We point out that Eq. (11) corresponds to the general format of the Ergun equation. However, we call it the Q-modified Ergun equation because the “constants” of $A = 268$ (approximately), and $B = \lambda/(2\pi\epsilon_0^3)$, have been modified from Ergun’s original constant values of 150 and 1.75 for A and B , respectively, and most importantly, B is not a constant, but rather is a function of both λ and ϵ_0 .

In Eq. (9), k_1 and k_2 are introduced. Definition of universal constant k_1 :

- Consider the control element of free space as a perfect sphere
- Volume of sphere $= \frac{4\pi}{3}r_h^3$
- Surface area of sphere (SA) $= 4\pi r_h^2$
- Cross-sectional area of sphere (CSA) $= \pi r_h^2$
- Ratio (SA)/(CSA) $= \frac{4\pi r_h^2}{\pi r_h^2} = 4$
- We define $r_h = 4$ as dynamic drag normalization coefficient
- We define $k_1 = \frac{\frac{4\pi}{3}r_h^3}{r_h} = \frac{4\pi}{3}r_h^2 = \frac{64\pi}{3} \approx 67$
- Thus, k_1 can be considered the wetted area relative to viscous contact and the relationship is a direct proportionality.

Definition of the universal constant k_2 :

- Consider the unit cylindrical fluid flow channel of radius 1
- Circumference (perimeter) of channel $= 2\pi$
- Impose a reciprocal relationship $= (1/2\pi)$
- We define $k_2 = \frac{1/2\pi}{r_h} = \frac{1}{8\pi} \approx 0.04$
- Justification is based upon the relationship between k_1 and k_2
- Control element $= \frac{4\pi}{3}r_h^3$
- Viscous control coefficient k_1 is a direct proportionality, so we divide by r_h which gives $\frac{4\pi}{3}r_h^2$
- Kinetic control coefficient k_2 is a reciprocal proportionality, so we divide by r_h which gives $\frac{1}{8\pi}$

III. METHODS

A. Experiment

To validate the Quinn fluid flow model (QFFM), we designed a continuous-flow apparatus capable of precise pressure drop and temperature measurements under controlled conditions. The system consists of a recirculating flow loop with calibrated instrumentation, including resistance temperature detector (RTD) sensors for temperature monitoring and pressure transducers for head loss quantification.²³ The apparatus (photo is presented in Fig. 1) consists of the following sub-blocks:

(1) Flow Loop System

- A recirculating fluid loop with calibrated pressure transducers for pressure drop measurements
- Multiple RTD (resistance temperature detector) sensors positioned along the pipe to monitor temperature at various locations
- A circular gear pump capable of generating fluid flow at pressures up to 200 psi (1.38 MPa)

(2) Control and Measurement Systems

- A computerized control system running LabVIEW software (National Instruments)
- Flow rate measurement using graduated cylinders and computer-synchronized stopwatch



FIG. 1. Photo of experimental setup (HMQ-4).

- Automated recording of temperature and pressure drop across test sections
- (3) Test Sections

- Teflon tubing (ID: 0.076 inches/1.93 mm) in three configurations: straight section: 241 cm length (fully extended); coiled sections: 7620 and 15 240 cm lengths
- Large plastic reservoirs containing temperature-controlled water (maintained at room temperature).

The operational protocol for the experiments is the following. The pump control voltage was set via software interface while simultaneously recording flow rate (via timed volumetric measurements), temperature (at multiple RTD positions), and pressure drop (across test sections) for each experimental run.

The packed column had the following dimensions: 3.85 mm inner diameter and 1032.67 mm length filled with 297 electro-polished stainless spherical steel particles (mean diameter $= 3.469 \text{ mm} \pm 0.5\%$ by caliper measurement). The monodisperse, smooth, nonporous particles yielded a bed porosity of 0.4557. Water was pumped through the system using an internal gear pump while pressure drops were measured with a high-precision transducer (accuracy $\pm 0.1\%$ FS). The set of experiments for these design parameters is named as HMQ-14 for further reference.

Other experimental data were adopted from the study of air flow through a structured bed published by Bai *et al.*²⁴ The experimental setup contained a vertically mounted square duct $20 \times 20 \text{ mm}$ filled

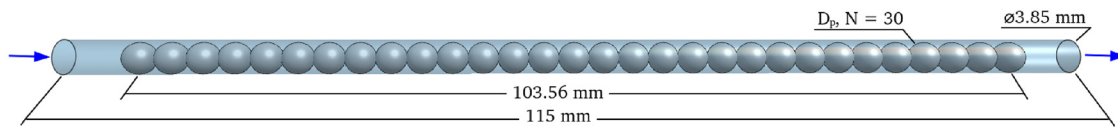


FIG. 2. Cylindrical domain.

with steel balls 10 mm in diameter. The packing comprised of 64 particles and had a height of 160 mm. A water manometer was used for measurement of pressure drop in the packed bed through two side ports. A compressed air feed was controlled with a manual valve and measured with a flow cell rotameter.

B. Numerical model

1. Computational domains

The model was to be validated against two sets of experimental data; thus, separate geometries were developed for each case: a cylindrical domain with spherical particles arranged in a one-dimensional close-packed chain and a square domain packed with spherical particles in 2×2 stacked layers.

The cylindrical domain was 115 mm long and had a diameter of 3.85 mm (Fig. 2). The particles of diameter $D_p = 3.469$ mm formed a linear array of $N = 30$ elements at a pitch of 3.468 mm (0.01% overlap). In accordance with experimental conditions, the particles were aligned to a side of the cylinder with an overlap of $0.347 \mu\text{m}$ (0.01% overlap). These overlaps were introduced to prevent singularities at contact points of curvilinear surfaces²⁵ without affecting flow dynamics. Entrance and exit regions of 6 mm length were used to stabilize the solution and control parameters at the boundaries.

The square domain was 260 mm long and had a cross section of 20×20 mm (Fig. 3). Sixty-four spheres of diameter $D_p = 10.0001$ mm each were arranged in an orthogonal array with a pitch of 10 mm. This ensured $0.1 \mu\text{m}$ (0.01%) overlap for the purpose stated above. The entrance and the exit regions were about 50 mm each.

Both domains contain structured grids of particles. This was an intentional choice among various packings with experimental data. The structured packings were preferred as they eliminate any uncertainties of particle allocations. This makes them perfect for validation purposes.

2. Mesh and its sensitivity analysis

The computational domain was discretized using an unstructured mesh comprised of nonuniform polyhedra cells with a growth rate of 1.2 (Fig. 4). Up to five prismatic layers with a growth rate of 1.5 were applied to all fluid–solid interfaces to enhance solution versatility in high gradient regions. First, layer thickness, growth rate, and number of layers varied across the mesh to ensure wall distance $y^+ \sim 1$ and minimize prismatic layers stepping.

Specification of spatial discretization is presented in Table I. Mesh naming consists of domain type (SD for the square domain and CD for the cylindrical one) and of maximum cell size in millimeters. The maximum cell size was set during mesh generation and correspond to a linear value that should be interpreted as a limiter for a cell area $S_{\max} = \Delta x^2$ at the surface mesh and a cell volume $V_{\max} = \Delta x^3$ at the volume mesh. Skewness is a cell quality characteristic showing how

severely the cell deviates from a perfectly regular polygon or polyhedron. Orthogonality is also a cell quality characteristic, but measuring cell's boundaries alignment with the bulk of the mesh. All considered meshes have good average quality characteristics (almost nonexistent skewness and close to unity orthogonality). There are some gruesome cells of marginal quality (maximum skewness and minimum orthogonality). However, these cells are quite rare ($<0.1\%$ of cells) and evenly distributed over the domains. Thus, it is safe to conclude that they are not a result of systematic generation error and are not affecting the solution noticeably.

Mesh dependency of simulation was evaluated on the square domain at a flow rate of 5.781 kg/s (see detailed description below). Maximum cell size Δx varied from 0.2 to 0.64 mm with local refinement factor up to 2.5 (i.e., 0.08...0.256 mm). For the considered case, Richardson extrapolation predicted pressure loss of 13.9 kPa (on theoretical mesh with element sizes $\Delta x \rightarrow 0$). With an acceptable error of 1% (green zone on Fig. 5), acceptable results were obtained on meshes with cell sizes up to 0.25 mm. Thus, $\Delta x = 0.25$ mm was selected as a default mesh size for the main simulations on square domain.

Trend analysis of mesh dependency confirmed results eligibility. An asymptotic trend was easily revealed, whereas other points were discarded as diverging. The shear stress (SST) model does not resolve eddies; thus, it relies on eddy viscosity to estimate stresses caused by turbulence. Since turbulent pulsations are being spatially averaged, mesh coarsening results in linear overestimation of boundary effect on fluid flow. Due to a clearly underestimating starting point (-12.1% of pressure loss at $\Delta x = 0.4$ mm), this growth compensated it. However, such compensation is uncontrollable and might yield significant discrepancies over the computational domain, considering its topological complexity.

The upper limit of cell size $\Delta x = 0.25$ mm for the cylindrical domain yielded mesh with much lower quality compared to the quadratic domain. That was because of a significantly smaller dimensions of the cylindrical domain. Thus, a further mesh refinement was applied, lowering the maximum size down to $\Delta x = 0.05$ mm. Mesh quality was significantly improved up to the level of the square domain (see Table I). A full mesh dependency analysis was not conducted for the cylindrical domain because simulation results were very similar on both meshes (discrepancy $<5\%$ without any trend). Thus, $\Delta x = 0.15$ mm was selected as a default mesh size for the main simulations on cylindrical domain.

3. Governing equations

The following basic formulation of Navier–Stokes equations was used:

$$\nabla \cdot \mathbf{u} = 0, \quad (12)$$

$$(\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu (\nabla^2 \mathbf{u}), \quad (13)$$

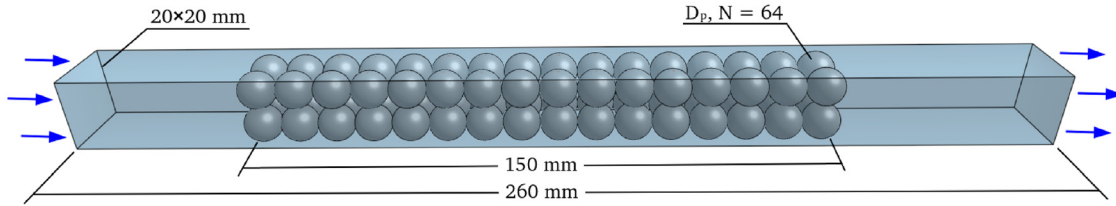


FIG. 3. Square domain.

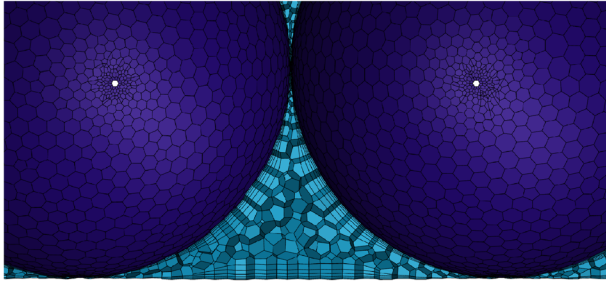


FIG. 4. Cross section of computational mesh.

$$R_{ij} = \mu_t \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) - \frac{2}{3} k \delta_{ij}. \quad (14)$$

Velocity divergence in continuity Eq. (12) was eliminated due to incompressibility of the working fluid (air). Momentum equation (13) has no time derivative due to steady-state formulation. There is also no gravity term, since the system was considered isothermal. The left-hand side of the momentum equation describes spatial distribution of the velocity field, whereas the right-hand side contains forces causing the flow: negative pressure gradient that drives the flow and viscous forces that smooth out velocity gradients and resist the motion.

TABLE I. Meshes specification.

Name	Cell size (mm)	Cell count	Skewness (avg/max)	Orthogonality (avg/min)
Cylindrical domain				
CD-0.25	0.25	0.34×10^6	0.16/0.97	0.83/0.03
CD-0.15	0.15	0.9×10^6	0.01/0.96	0.99/0.01
CD-0.05	0.05	13.16×10^6	0.03/0.97	0.97/0.03
Square domain				
SD-0.64	0.64	1.02×10^6	0.11/0.85	0.89/0.15
SD-0.5	0.5	1.72×10^6	0.02/0.88	0.98/0.13
SD-0.45	0.45	2.31×10^6	0.04/0.84	0.95/0.13
SD-0.4	0.4	2.93×10^6	0.06/0.86	0.94/0.14
SD-0.35	0.35	4.6×10^6	0.05/0.81	0.94/0.11
SD-0.3	0.3	6.1×10^6	0.04/0.88	0.96/0.12
SD-0.25	0.25	10.05×10^6	0.01/0.97	0.99/0.11
SD-0.2	0.2	18.02×10^6	0.03/0.87	0.97/0.14

Reynolds stress [Eq. (14)] was calculated with turbulent viscosity, which was estimated with $k - \omega$ SST turbulence model proposed by F. Mentor in the early 1990s:²⁶

$$\mu_t = \frac{0.31k\rho}{\max\left(0.31\omega, F_2 \frac{\partial u}{\partial y}\right)}, \quad (15)$$

$$F_2 = \tanh\left(\left(\max\left(\frac{2\sqrt{k}}{0.09\omega y}, \frac{500\mu}{\rho y^2 \omega}\right)\right)^2\right). \quad (16)$$

The term $-\frac{2}{3}k\delta_{ij}$ in Eq. (14) utilizes Kronecker delta to ensure the trace of the Reynolds stress tensor R_{ij} is zero for incompressible flows, separating volumetric effects from shear effects. The turbulent viscosity μ_t is used to account for an intensification of molecular diffusion by turbulent eddies; thus, the effective viscosity is $\mu_{eff} = \mu + \mu_t$. The shear-stress transport model calculates turbulent viscosity μ_t using a semi-empirical correlation that blends features of the $k - \omega$ (near-wall) and $k - \varepsilon$ (far-field) models. This blending is controlled by the function F_2 , which ensures a smooth transition between the two formulations.

The simulation of fluid flow in a packed bed of spherical particles involves complex flow regimes, including laminar, transitional, and turbulent regions, particularly at higher Reynolds numbers. The primary objective is to accurately predict pressure losses, which depend on resolving near-wall behavior, flow separation, and adverse pressure gradients. Among the available turbulence models, the shear stress transport (SST) $k - \omega$ model is well-suited for this application, offering advantages over alternatives such as large eddy simulation (LES) and standard $k - \varepsilon$ models.

LES is prohibitively expensive for industrial-scale packed bed simulations due to the need for extremely fine grids and transient resolution. Since the primary goal of the study is pressure loss estimation (not detailed turbulence structures), a well-calibrated RANS model like $k - \omega$ SST was considered to be more efficient.

The standard $k - \varepsilon$ models overpredict turbulent viscosity and perform poorly in transitional flows with strong separation or curvature, which are prevalent in packed beds at moderate Reynolds numbers. These models are calibrated for high-Reynolds flows with minimal wall interactions; thus, they utilize wall functions with very coarse meshes near the walls ($y^+ > 30$). However, the close packing of the particles requires much finer meshes in near wall regions, especially near the prospective contact points.

4. Boundary conditions

At the inlet, a uniform velocity profile was set with Dirichlet boundary condition: $u(x) = U_0 \vec{n}$. A flat velocity profile was used

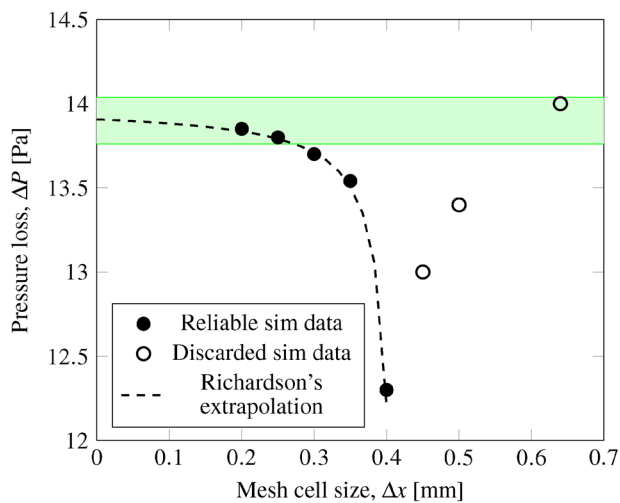


FIG. 5. Mesh dependency analysis.

instead of a fully developed one because the contribution of wall shear stress (at entrance and exit regions) to the overall pressure loss is negligible compared to that of the packed bed. Velocity was derived from mass flow rates available in experimental data.^{20,24} Turbulence parameters of the inlet flow were set with turbulent intensity $I = \frac{5.5}{Re^{0.0407}}$ and turbulent viscosity ratio $R_t = \frac{0.09 \rho k^2}{\epsilon \mu}$, where Re is Reynolds number, ρ and μ are fluid density and viscosity, $k = 1.5(U_0 I)^2$ is turbulent kinetic energy, and $\epsilon = \frac{0.164 k^{1.5}}{L_m}$ is turbulent dissipation rate. The turbulent length scale $L_m = 0.4 \frac{0.382 x}{Re^{0.2}}$ was calculated with coordinate-related Reynolds number $Re_x = \frac{U_0 x \rho}{\mu}$, where x is the entrance region length. Detailed specification of inlet boundary conditions is listed in Table II.

At the outlet, a surface-normal zero gradient of pressure ($\partial p / \partial \vec{n}$) with $p_{\text{gauge}} = 0$ was set due to flow incompressibility. Additionally, the backflow was restricted to stabilize the solution. Fluid–solid interfaces (i.e., channel and particles walls) were set with no-slip boundary condition with artificial roughness of 45 μm for the square domain. As particles were polished in experiments with cylindrical domain, artificial roughness was not applied to them.

5. Fluid properties

Assumption of incompressible and isothermal flow reduced operational properties of the working fluids to density and viscosity. For water in the cylindrical domain, they were set to 1000 kg/m^3 and 1 $\text{mPa} \cdot \text{s}$, whereas for air in the square domain—to 1.194 kg/m^3 and $1.81 \times 10^{-5} \text{ Pa} \cdot \text{s}$.

C. Validation protocol

To ensure an accurate and complete solution with the numerical model, convergence checks were applied during each simulation run. Control parameter (pressure drop) stabilization was continuously monitored over a moving average of 10 iterations with a stopping criterion of 0.1%

TABLE II. Inlet boundary conditions.

G^m (g/s)	I (%)	R_t
Cylindrical domain		
0.000 165	...	...
0.056 201	4.882	0.037
0.112 237	4.747	0.071
0.168 273	4.669	0.105
0.224 309	4.615	0.138
0.280 345	4.573	0.171
0.336 381	4.540	0.204
0.392 417	4.511	0.237
0.448 453	4.487	0.269
0.504 489	4.465	0.301
0.560 525	4.446	0.333
Square domain		
2.892	3.815	12.266
4.914	3.734	20.397
5.781	3.709	23.838
7.342	3.673	29.982

$$\left| \sum_{i=10}^i \frac{p_{in} - p_{out}}{10} - \sum_{i=11}^{i-1} \frac{p_m - p_{out}}{10} \right| < 0.001(p_{in} - p_{out}). \quad (17)$$

Additionally, numerical residuals were continuously monitored with the following stopping criteria: $E_{\text{continuity}} = 0.01$, $E_{\text{velocity}} = 10^{-5}$, and $E_{\text{pressure}} = 10^{-6}$.

Pressure drop was calculated as a difference between inlet and outlet pressures. However, actual differential pressure measurements were made through side-mounted ports.^{20,24} Thus, both direct and indirect (via side ports) approaches were evaluated with CFD. Preliminary simulations showed no noticeable difference in pressure loss regardless of capturing method. Thus, a more straightforward approach (direct) was used in the main simulations.

IV. RESULTS

A. Water flow

To evaluate the predictive accuracy of the Quinn fluid flow model (QFFM, Quinn's law), we analyzed the logarithmic relationship between the normalized dimensionless pressure gradient (P_Q) and normalized dimensionless fluid current (C_Q) with our experimental results (HMQ-14) and the results from other works:

- The Oregon pipe flow study²⁷
- Giddings' packed bed measurements²²
- Nikuradze's smooth pipe data²⁸
- The Princeton superpipe experiments^{29,30}
- Farkas' validation study³¹
- Ergun's classical fluidization work⁷

Figure 6 compares experimental data from multiple sources for the normalized dimensionless pressure gradient (P_Q) vs normalized dimensionless fluid current (C_Q). The log–log representation shows excellent agreement with QFFM predictions spanning laminar, transient, and turbulent flow regimes.

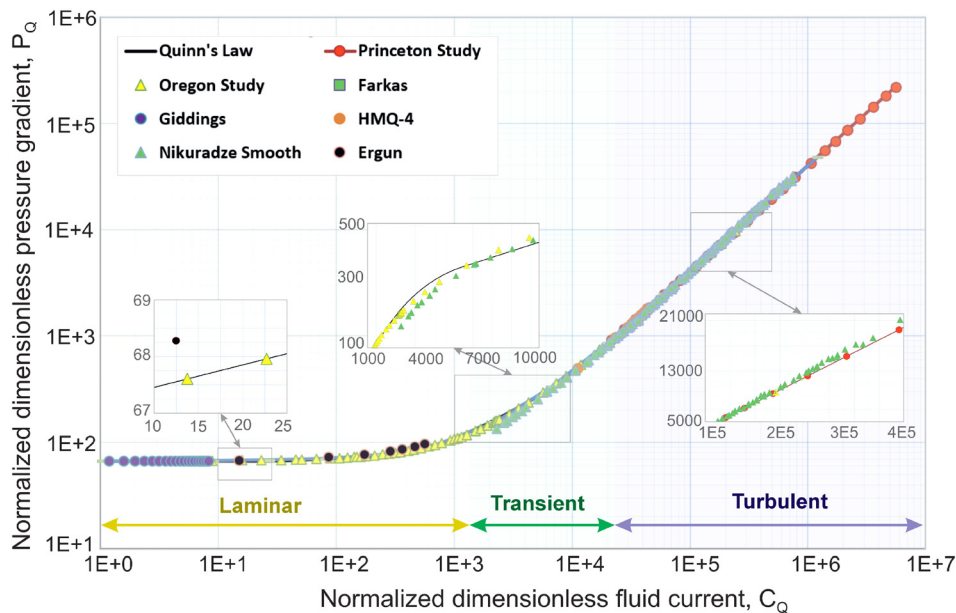


FIG. 6. Dimensionless plot of viscous friction factor P_Q vs wall-effect normalized fluid current C_Q according to the different sources: Oregon study;²⁷ Giddings;²² Nikuradze Smooth;²⁸ Princeton study;^{29,30} Farkas;³¹ Ergun;⁷ HMQ-4—our experimental data.

The QFFM was further validated against numerical simulations performed in this study as shown in Fig. 7. An overall agreement of these models is sufficient to confirm QFFM accuracy. The root mean squared discrepancy was below 1%, whereas the absolute maximum reached 3.7%. However, obtained trends of pressure loss are consistent.

Simulation results for the cylindrical domain showed a uniform streamwise gradient of pressure (see Fig. 8). The velocity profile was consistent along the domain axis, with peaks above the spheres in the widest pathway. Tangency of the particles with the bottom part of the cylinder resulted in a stagnant zone that spanned over approximately 1/4th sector of the domain. Even at the highest flow rates of the considered range, turbulent intensity was moderate (below 4.8%) throughout the domain. This emphasizes a moderate contribution of turbulent viscosity in the momentum formulation [see Eq. (14)] and lowers an uncertainty rate of predictions.

It should be noted that numerical results appeared to be very sensitive to the particles' alignment inside the cylinder. The first approach utilized centered alignment (i.e., coincidence of spheres' centers with cylinder axis) and yielded noticeably higher pressure loss with an increase in about 15.8% over the full range of flow rates. This might be explained with a shear stress forces that are proportional to the ratio of perimeter and cross section area for each pathway. From the topology perspective, particles alignment does not affect that ratio. However, particles' displacement creates a crescent-shaped pathway with a much higher ratio, thus lowering shear effects. The detailed mechanism of the phenomenon is to be elaborated in the future studies.

B. Air flow

Simulation results for the square domain showed expected distributions of pressure and velocity fields. The pressure gradient was uniform over the packed bed in the streamwise direction (see Fig. 9).

The velocity field was also uniform in streamwise direction, with a symmetrical distribution over lateral sections of the packing. An overall flow stabilized at the fifth layer of particles, with a central interstitial pathway being the last to develop (see velocity streamlines on Fig. 9). This central flow had the highest velocity (up to 108 m/s at $G^m = 7.342$ g/s) since it had the largest hydraulic cross section $A_h = D_p^2(1 - \pi/6)$ that ranges from $A_{h,min} = D_p^2(1 - \pi/4)$ to $A_{h,max} \approx D_p^2$. At a flow rate $G^m = 7.342$ g/s, velocity within the central pathway reaches 108 m/s that is five times of inlet velocity. This holds for the whole considered range of flow rates, with a minor variation of less than 1.4%.

The lateral distribution of the velocity field is highly nonuniform, but orthogonally symmetrical (see Fig. 10). At the minimum hydraulic

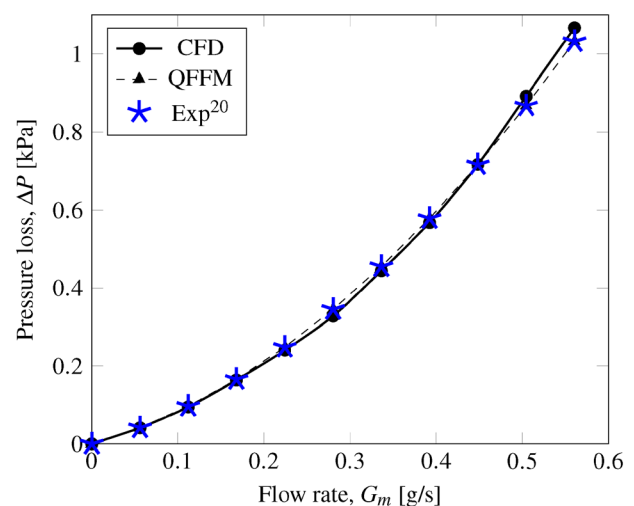
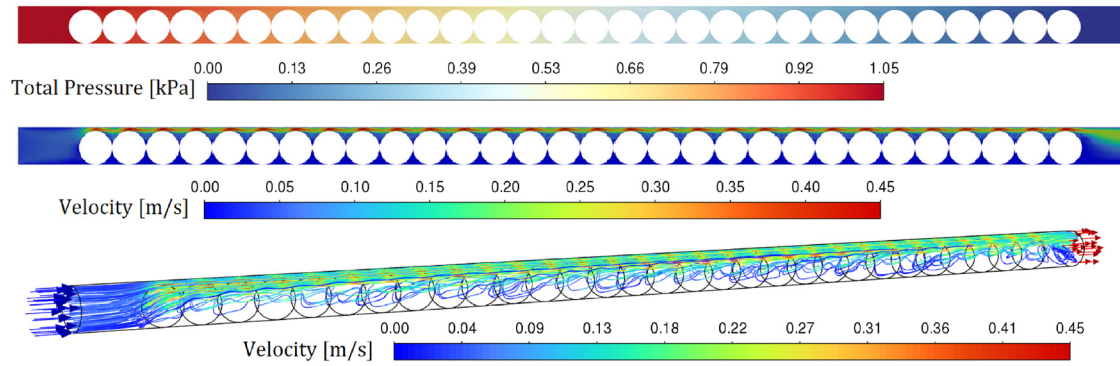


FIG. 7. Validation of the cylindrical domain simulation results.

FIG. 8. Simulation results for the cylindrical domain at $G_m = 0.56$ g/s.

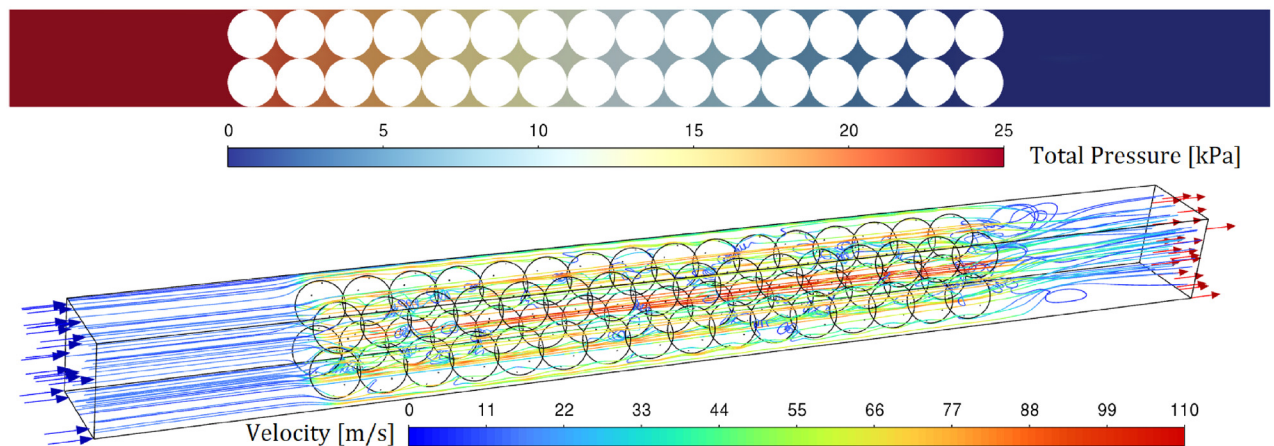
cross section (i.e., near particles' equators), the average velocity is relatively high (71.5 m/s at $G^m = 7.342$ g/s). However, velocity variation is minimal: area-weighted uniformity index $\gamma_a = 1 - \frac{\sum (\psi_i - \psi_{avg}) A_i}{2\psi_{avg} A_{cs}} = 0.9$, where ψ_i and ψ_{avg} are facet and cross-sectional averaged value, A_i and A_{cs} are facet and cross-sectional areas. At the maximum hydraulic cross section (i.e., near particles' poles), the average velocity is much lower (23.4 m/s at $G^m = 7.342$ g/s), whereas its variation is higher ($\gamma_a = 0.61$).

Obtained lateral sections (Fig. 10) showed a highly symmetrical pattern of flow distribution, with a throughput proportional to the averaged hydraulic cross section A_h of the pathways. The correlation is nonlinear and is a subject of the future work.

Simulation results were in a good agreement with adopted experimental data²⁴ (Fig. 11). The average discrepancy over four data points was less than 0.55 kPa (2.2% of the full scale). However, CFD simulation overestimates pressure loss at medium and low flow rates compared to QFFM, that on the contrary, underestimates it at higher flow rates. As both models provide smooth correlations between flow rate and pressure loss, the mentioned discrepancies were considered indecisive.

V. CONCLUSION

This study presented a comprehensive experimental and numerical validation of the Quinn fluid flow model (QFFM) for predicting pressure drop in packed beds. The QFFM, derived from first principles, establishes a universal linear relationship between the normalized dimensionless pressure gradient (P_Q) and fluid current (C_Q), expressed as $P_Q = k_1 + k_2 C_Q$, where $k_1 \approx 67$ (viscous constant) and $k_2 \approx 0.04$ (kinetic constant). Key findings include validation across flow regimes, where the QFFM demonstrated good agreement with experimental data spanning laminar, transient, and turbulent flow regimes. For water flow in cylindrical packed beds (porosity $\varepsilon = 0.488$), the model predicted pressure drops with a root-mean-square discrepancy of $<1\%$ and a maximum absolute error of 3.7%. For air flow in structured square beds, the QFFM matched experimental measurements with an average deviation of 0.55 kPa (2.2% of full scale), outperforming the Ergun equation, which exhibited deviations up to 15% under similar conditions. Numerical study was done via CFD simulations, with mesh sensitivity analysis showing convergence at cell sizes ≤ 0.25 mm (error $< 1\%$). Turbulent intensity remained moderate ($< 4.8\%$) even at high flow rates ($G_m = 7.342$ g/s), underscoring the QFFM's applicability across flow regimes without requiring empirical adjustments for

FIG. 9. Simulation results for the square domain at $G_m = 7.342$ g/s.

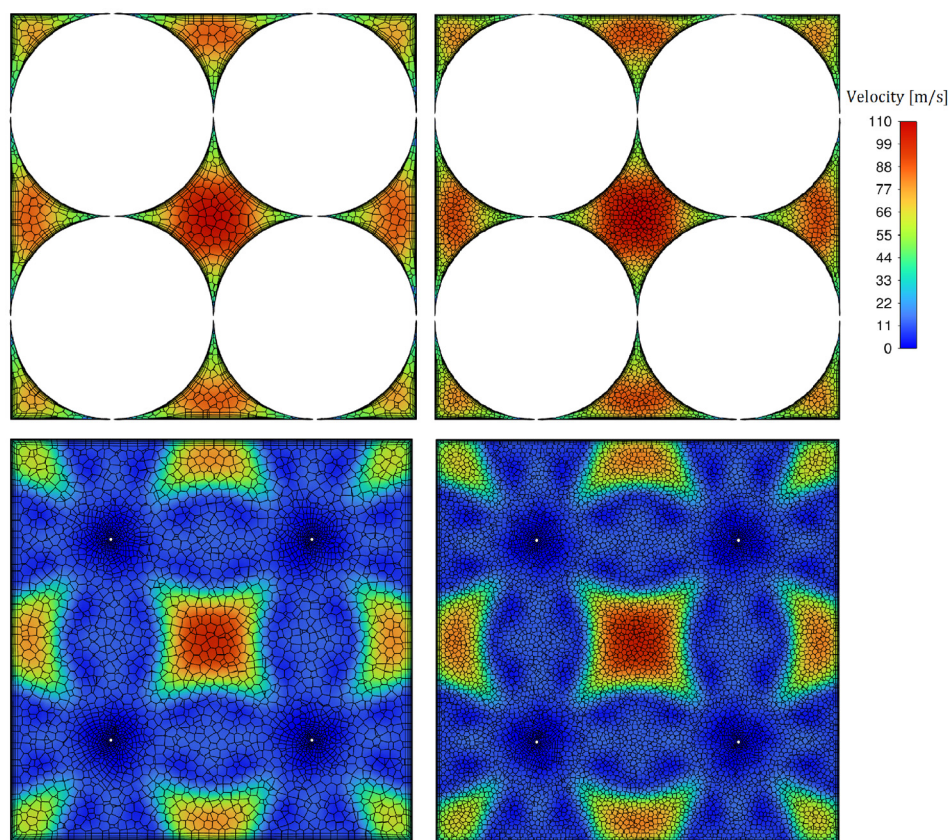


FIG. 10. Velocity profiles in cross sections of the square domain at $G_m = 7.342$ g/s on meshes with $\Delta x = 0.5$ mm (left) and $\Delta x = 0.25$ mm (right).

turbulence. The QFFM resolved limitations of traditional models (e.g., Ergun equation) by inherently accounting for tortuosity and micro-scale flow heterogeneity. For low D/d ratios (e.g., 1.1), the QFFM reduced prediction errors by $>50\%$ compared to the Ergun equation,

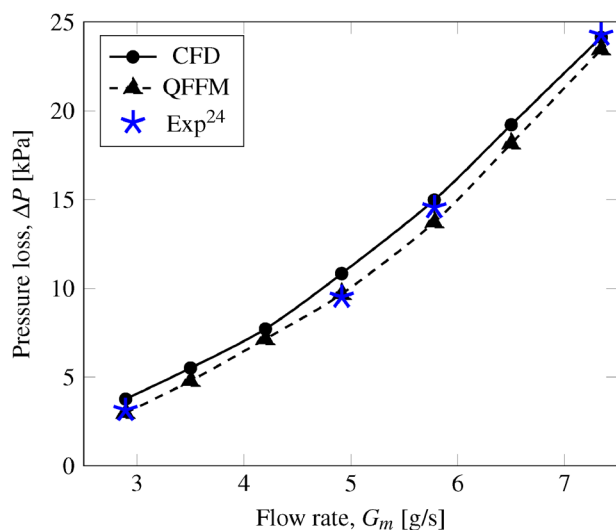


FIG. 11. Validation of the square domain simulation results.

which struggles with such geometries. The QFFM's unified framework—applicable to both empty conduits and packed beds—eliminated the need for case-specific constants, as demonstrated by its consistency across disparate datasets. Limitations of the current validation include its restriction to monodisperse spherical particles and single-phase flows, suggesting future studies should explore polydisperse packings and multiphase systems. Extending the QFFM to industrial-scale reactors (e.g., pseudo-homogeneous models) and reacting flows could further broaden its utility. In summary, the QFFM provides a clear, physics-based alternative to semi-empirical correlations, offering engineers a reliable tool for optimizing packed-bed systems, with validation across diverse geometries and flow conditions underscoring its potential to advance fluid dynamics modeling in both academic and industrial settings.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Authors Contribution

M. Nikitin: Data curation (equal); Investigation (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal). **H. M. Quinn:** Conceptualization (equal); Data curation (equal); Investigation (equal); Methodology (equal); Writing – original draft (equal); Writing – review & editing (equal). **D. Pashchenko:**

Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (equal); Software (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

NOMENCLATURE

Abbreviations

HQC Hypothetical Q channel
QFFM Quinn fluid flow model

Greek symbols

γ_a Area-weighted uniformity index
 δ Porosity normalization coefficient
 ε Q-porosity function
 ε_i Conduit internal volume fraction
 ε_p Particle porosity
 ε_{sk} Conduit particle skeletal volume fraction
 ε_t Conduit total volume fraction
 ε_0 Conduit external volume fraction
 Θ Dimensionless permeability
 λ Wall normalization coefficient
 μ_s Fluid superficial velocity
 ν Fluid absolute viscosity
 ρ_f Fluid density

Roman symbols

A Viscous constant
 A_h Hydraulic cross-sectional area
 B Kinetic constant
 C_Q Fluid current for fluid flow in closed conduits
 d_c Channel diameter
 d_p Average spherical particle diameter equivalent
 D_p Particle diameter
 f_v Viscous friction factor
 g Acceleration due to gravity
 G^m Fluid mass flow rate
 I Turbulent intensity
 k_1 Universal viscous constant
 k_2 Universal kinetic constant
 L Channel length
 L_m Turbulent length scale
 n_k Kinetic hydraulic force
 n_p Number of particles
 n_{pq} Volume of the conduit expressed in terms of number of particles
 n_v Viscous hydraulic force
 P_Q Normalized dimensionless pressure gradient
 q Volumetric flow rate
 Q_N Instantaneous fluid current

r_h Fluid drag normalization coefficient
 R_t Turbulent viscosity ratio
 Re_m Modified Reynolds number
 y^+ Dimensionless wall distance
 ΔP Pressure loss
 Δx Mesh size

REFERENCES

- ¹A. De Wasch and G. Froment, "Heat transfer in packed beds," *Chem. Eng. Sci.* **27**, 567–576 (1972).
- ²D. Iranshahi, A. Golrokh, E. Pourazadi, S. Saedi, and F. Gallucci, "Progress in spherical packed-bed reactors: Opportunities for refineries and chemical industries," *Chem. Eng. Process.-Process Intensif.* **132**, 16–24 (2018).
- ³D. Wen and Y. Ding, "Heat transfer of gas flow through a packed bed," *Chem. Eng. Sci.* **61**, 3532–3542 (2006).
- ⁴R. Hicks, "Pressure drop in packed beds of spheres," *Ind. Eng. Chem. Fundam.* **9**, 500–502 (1970).
- ⁵K. Allen, T. Von Backström, and D. Kröger, "Packed bed pressure drop dependence on particle shape, size distribution, packing arrangement and roughness," *Powder Technol.* **246**, 590–600 (2013).
- ⁶S. Afandizadeh and E. Foumeny, "Design of packed bed reactors: Guides to catalyst shape, size, and loading selection," *Appl. Therm. Eng.* **21**, 669–682 (2001).
- ⁷S. Ergun, "Fluid flow through packed columns," *Chem. Eng. Prog.* **48**, 89 (1952).
- ⁸P. C. Carman, "Fluid flow through granular beds," *Trans. Inst. Chem. Eng. London* **15**, 150–156 (1937).
- ⁹R. B. Bird, E. N. Lightfoot, and W. E. Stewart, "Notes on transport phenomena. Transport phenomena," in *A Rewritten and Enlarged Edition of Notes on Transport Phenomena*, edited by R. B. Bird, Warren E. Stewart, and Edwin N. Lightfoot (John Wiley & Sons, New York, London, 1960).
- ¹⁰I. Macdonald, M. El-Sayed, K. Mow, and F. Dullien, "Flow through porous media-the Ergun equation revisited," *Ind. Eng. Chem. Fundam.* **18**, 199–208 (1979).
- ¹¹X. Wu and T. Hibiki, "Pressure drop coefficient correlation for forced convective flows in packed beds of spherical particles," *Int. J. Heat Mass Transfer* **240**, 126620 (2025).
- ¹²Z. Guo, Z. Sun, N. Zhang, and M. Ding, "Influence of confining wall on pressure drop and particle-to-fluid heat transfer in packed beds with small d/d ratios under high Reynolds number," *Chem. Eng. Sci.* **209**, 115200 (2019).
- ¹³V. Papkov, N. Shadymov, and D. Pashchenko, "Gas flow through a packed bed with low tube-to-particle diameter ratio: Effect of pellet roughness," *Phys. Fluids* **36**, 027127 (2024).
- ¹⁴H. Zhang, J. Xu, L. Guo, L. Gao, J. Ye, and W. Ge, "Coupling numerical simulation and artificial intelligence prediction: A computational fluid dynamics-discrete element method and deep learning approach to gas-solid flows," *Phys. Fluids* **37**, 053330 (2024).
- ¹⁵A. Jafari, P. Zamankhan, S. Mousavi, and K. Pietarinen, "Modeling and CFD simulation of flow behavior and dispersivity through randomly packed bed reactors," *Chem. Eng. J.* **144**, 476–482 (2008).
- ¹⁶D. Lanade and Y. Hassan, "Numerical study of convective heat transfer in randomly packed beds of low aspect ratios," *Phys. Fluids* **36**, 105120 (2024).
- ¹⁷N. Ahmed and D. K. Sunada, "Nonlinear flow in porous media," *J. Hydraul. Div.* **95**, 1847–1858 (1969).
- ¹⁸A. Koekemoer and A. Luckos, "Effect of material type and particle size distribution on pressure drop in packed beds of large particles: Extending the Ergun equation," *Fuel* **158**, 232–238 (2015).
- ¹⁹T. Buchwald, G. Schmandra, L. Schützenmeister, T. Fraszczak, T. Mütze, and U. Peuker, "Gaseous flow through coarse granular beds: The role of specific surface area," *Powder Technol.* **366**, 821–831 (2020).
- ²⁰H. M. Quinn, "Quinn's law of fluid dynamics pressure-driven fluid flow through closed conduits," *Fluid Mech.* **5**, 39–71 (2020).
- ²¹P. Forchheimer, "Wasserbewegung durch boden," *Z. Ver. Dtsch. Ing.* **45**, 1781–1788 (1901).
- ²²J. C. Giddings, "Dynamics of chromatography: Part I: Principles and theory," in *Dynamics of Chromatography* (Marcel Dekker, New York, 1965), Vol. 1.

- ²³H. M. Quinn, “Quinn’s law of fluid dynamics: Supplement# 1 Nikuradze’s inflection profile revisited,” *Fluid Mech.* **6**, 1–14 (2020).
- ²⁴H. Bai, J. Theuerkauf, P. A. Gillis, and P. M. Witt, “A coupled DEM and CFD simulation of flow field and pressure drop in fixed bed reactor with randomly packed catalyst particles,” *Ind. Eng. Chem. Res.* **48**, 4060–4074 (2009).
- ²⁵A. G. Dixon, M. Nijemeisland, and E. H. Stitt, “Systematic mesh development for 3D CFD simulation of fixed beds: Contact points study,” *Comput. Chem. Eng.* **48**, 135–153 (2013).
- ²⁶F. Menter, “Zonal two equation kw turbulence models for aerodynamic flows,” in *23rd Fluid Dynamics, Plasmadynamics, and Lasers Conference* (AIAA, Orlando, 1993), p. 2906.
- ²⁷C. J. Swanson, B. Julian, G. G. Ihas, and R. J. Donnelly, “Pipe flow measurements over a wide range of Reynolds numbers using liquid helium and various gases,” *J. Fluid Mech.* **461**, 51–60 (2002).
- ²⁸J. Nikuradse, “Stromungsgesetze rauhen rohren,” *vdi-forschungsheft* **361**, 1 (1933).
- ²⁹B. McKeon, C. Swanson, M. Zagarola, R. Donnelly, and A. J. Smits, “Friction factors for smooth pipe flow,” *J. Fluid Mech.* **511**, 41–44 (2004).
- ³⁰B. McKeon, M. Zagarola, and A. Smits, “A new friction factor relationship for fully developed pipe flow,” *J. Fluid Mech.* **538**, 429–443 (2005).
- ³¹T. Farkas, G. Zhong, and G. Guiochon, “Validity of Darcy’s law at low flow-rates in liquid chromatography,” *J. Chromatogr. A* **849**, 35–43 (1999).